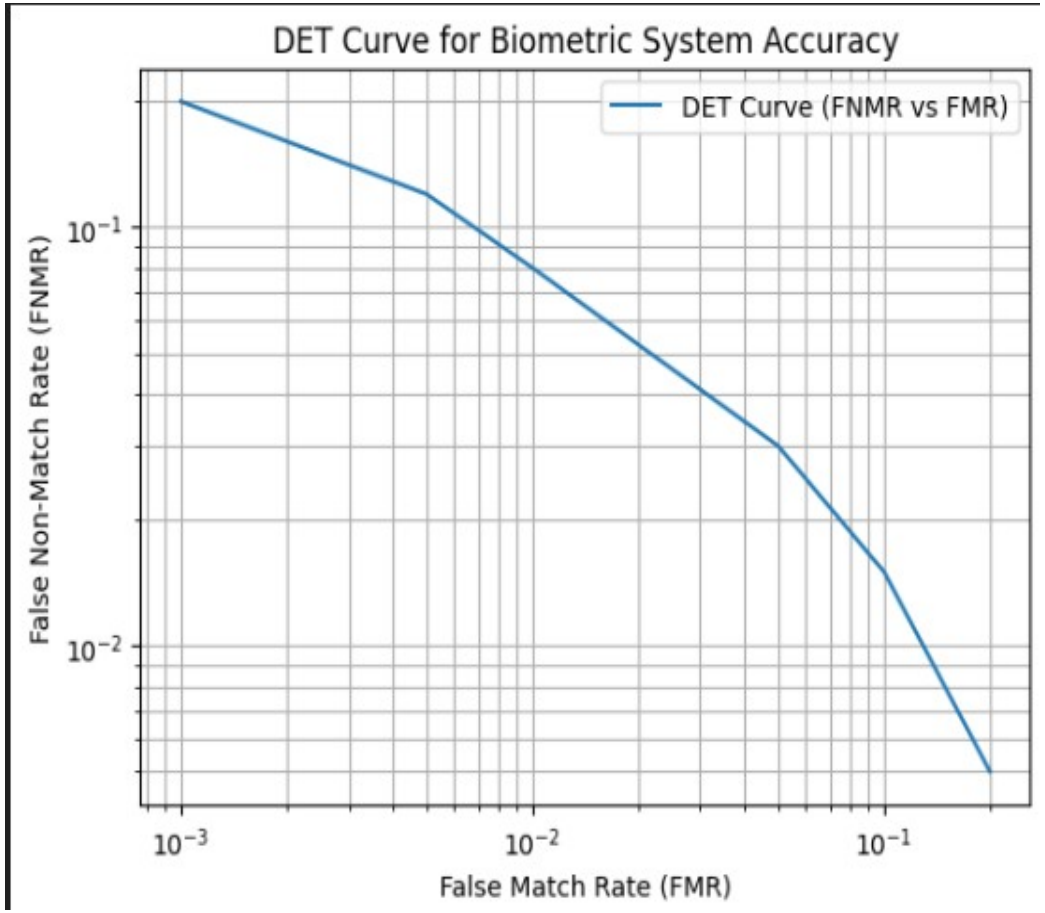
Accuracy in Biometric Systems

FMR, FNMR, FTR & Derived Metrics

DET Curve (Detection Error Trade-off)

- **DET Curve:**
  - Shows the trade-off between **security errors (FMR)** and **usability errors (FNMR)**
  - Both axes are in **logarithmic scale** to clearly visualize small error rates
- **Better biometric system:**
  - Curve lies **closer to the origin (0,0)**
  - Indicates **low FMR and low FNMR simultaneously**
- **Interpretation:**
  - Left movement → better security (lower FMR)
  - Downward movement → better user convenience (lower FNMR)



DET vs ROC

Feature	ROC Curve	DET Curve
Axes	TMR vs FMR	FNMR vs FMR
Scale	Linear	Logarithmic
Better	Top-left corner	Origin (bottom-left)
Use	General comparison	Low error rate comparison

Failure to Enroll (FTE) & Failure to Acquire (FTA)

- **FTE (Definition):** Users who cannot be enrolled in the system

FTE = (Failed_Enrollments / Total_Enrollment_Attempts)

- **Causes:**
  - Poor quality biometric
  - Physical disabilities
  - Sensor incompatibility

- **FTA (Definition):** System cannot capture biometric sample

FTA = (Failed_Acquisitions / Total_Acquisition_Attempts)

- **Causes:**
  - User interaction errors
  - Environmental conditions
  - Sensor malfunction

Important: FTE and FTA represent system failures before matching occurs. They directly impact system usability and user acceptance.

Accuracy & Error Rate

- **Accuracy:** Overall correctness of the biometric system.

  Accuracy = (TM + TNM) / (TM + TNM + FM + FNM)

  Accuracy = Correct Decisions / Total Attempts

- **Error Rate**

  Error Rate = 1 – Accuracy

  Error Rate = (FM + FNM) / Total

- **Why Not Just Use Accuracy?**

  Accuracy can be misleading when data is imbalanced

Example: If 99% attempts are genuine, system that accepts everyone has 99% accuracy but terrible security!

Example: System with TM=950, TNM=9800, FM=200, FNM=50
Accuracy = (950 + 9800) / 11000 = 0.977 (97.7%)

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Example: System with TM=950, TNM=9800, FM=200, FNM=50
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The system makes **mostly correct decisions**

Still, **200 imposters were accepted (FM)**

50 genuine users were rejected (FNM)

Hence, even with **97.7% accuracy**, error rates like **FMR and FNMR** must be analyzed.

Precision and Recall

Precision: Positive Predictive Value

- **Precision = TM / (TM + FM)**

Of all matches declared, what fraction are correct?

- **Focus:** Reliability of match decisions

Recall: Same as TMR/TAR

- **Recall = TM / (TM + FNM)**

Of all genuine users, what fraction are accepted?

- **Focus:** Coverage of genuine users

When to Use

- **High Precision Needed:** When false matches are costly (security applications)
- **High Recall Needed:** When missing genuine users is problematic (user convenience)

F-Score (F-Measure)

- **F1-Score:** Harmonic mean of Precision and Recall

  F1 = 2 × (Precision × Recall) / (Precision + Recall)

  F1 = 2×TM / (2×TM + FM + FNM)

Generalized F-Score

- **$F_\beta = (1 + \beta^2) \times (\text{Precision} \times \text{Recall}) / (\beta^2 \times \text{Precision} + \text{Recall})$**
  - **$\beta = 1$:** F1-score (equal weight to precision and recall)
  - **$\beta < 1$:** More weight to precision
  - **$\beta > 1$:** More weight to recall (e.g., F2-score with $\beta=2$)

- **Use Case:** F-score provides a single metric balancing precision and recall, useful when you need to consider both false matches and false non-matches.

d-prime (d') - Decidability

- **d-prime (d'):** Statistical measure of separation between genuine and impostor distributions
- **$d' = |\mu_genuine - \mu_impostor| / \sqrt{((\sigma^2_genuine + \sigma^2_impostor) / 2)}$**

Components

μ_genuine: Mean of genuine match scores

μ_impostor: Mean of impostor match scores

σ_genuine: Standard deviation of genuine scores

σ_impostor: Standard deviation of impostor scores

Interpretation

d' Value	Interpretation	System Quality
$d' < 1$	High overlap between distributions	**Poor**
$1 \leq d' < 2$	Moderate separation	**Fair**
$2 \leq d' < 3$	Good separation	**Good**
$d' \geq 3$	Excellent separation	**Excellent**

Decidability Index

- **Decidability Index (D):** Alternative measure of distribution separation

$$D = |\mu_{genuine} - \mu_{impostor}| / \sqrt{(\sigma^2_{genuine} + \sigma^2_{impostor})}$$

d' vs Decidability Index

Aspect	d-prime (d')	Decidability (D)
Denominator	Average variance	Sum of variances
Sensitivity	Less sensitive to unequal variances	More sensitive to outliers
Usage	More common in research	Alternative metric

Practical Note: Both metrics measure how well a matcher can distinguish between genuine and impostor attempts. Higher values indicate better discrimination capability.

Area Under Curve (AUC)

AUC - Probability that a randomly chosen genuine score is higher than a randomly chosen impostor score.

$$0 \leq AUC \leq 1$$

Interpretation

- **AUC = 1.0:** Perfect classifier (100% discrimination)
- **AUC = 0.9-0.99:** Excellent performance
- **AUC = 0.8-0.89:** Good performance
- **AUC = 0.7-0.79:** Fair performance
- **AUC = 0.5:** Random guessing (no discrimination)
- **AUC < 0.5:** Worse than random (system is inverted)

Advantages of AUC

- Threshold-independent metric
- Single number for overall performance
- Accounts for all possible operating points
- Useful for comparing different systems

Practical Calculation Example

- A fingerprint system tested with 1000 genuine attempts and 10,000 impostor attempts

- **Results at threshold $\tau = 0.6$**
  - Genuine attempts: 980 matched, 20 rejected
  - Impostor attempts: 50 matched, 9,950 rejected

Calculate the metrics:

TM = 980 (genuine accepted)
FNM = 20 (genuine rejected)
FM = 50 (impostor accepted)
TNM = 9,950 (impostor rejected)

FNMR = 20/1000 = 0.02 = 2%
FMR = 50/10,000 = 0.005 = 0.5%
TMR (TAR) = 980/1000 = 0.98 = 98%
TNMR (TRR) = 9,950/10,000 = 0.995 = 99.5%

Accuracy = (980 + 9,950) / 11,000 = 0.9936 = 99.36%
Precision = 980 / (980 + 50) = 0.9515 = 95.15%
F1-Score = 2 × (0.9515 × 0.98) / (0.9515 + 0.98) = 0.9657 = 96.57%

Comparing Biometric Systems

System	EER	FMR @ FNMR= 1%	d'	AUC	Ranking
System A	0.5%	0.01%	3.2	0.998	**Best**
System B	1.2%	0.1%	2.8	0.992	**Good**
System C	2.5%	0.5%	2.1	0.975	**Fair**

- **Comparison Guidelines**
  - **Use Multiple Metrics:** Don't rely on a single metric
  - **Consider Application:** High-security vs. convenience requirements
  - **Look at Operating Points:** Check performance at specific FMR/FNMR values
  - **Test Conditions:** Ensure fair comparison (same dataset, environment)
  - **Population Coverage:** Consider FTE and FTA rates

Testing Methodology for Metrics

1. Data Collection

- **Genuine Attempts:** Multiple samples from each enrolled user
- **Impostor Attempts:** Cross-comparisons between different users
- **Sample Size:** Minimum 100 users, 10 samples each recommended

2. Test Protocol (ISO/IEC 19795)

- **Technology Testing:** Lab environment, controlled conditions
  Scenario Testing: Simulated operational environment
  Operational Testing: Real-world deployment

3. Statistical Significance

- Calculate confidence intervals (typically 95%)
- Report standard errors for FMR and FNMR
- Ensure sufficient samples for rare events (low FMR)

- **Confidence Interval = Metric ± z × √(Metric × (1-Metric) / n)**

Summary

Essential Metrics

- FMR (Security)
- FNMR (Usability)
- EER (Benchmark)
- FTE/FTA (Coverage)

Advanced Metrics

- ROC/DET curves
- d-prime (Separation)
- AUC (Overall)
- F-Score (Balance)